

Primary-Side Control of Bipolar Field Excitation System for Pole-Changing Electrically Excited Synchronous Machines

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Abstract—This letter presents a novel bipolar field excitation system for pole-changing electrically excited synchronous machines. Conventional wireless power transfer-based systems realize bipolar field current through receiver-side active converters and wireless gate-signal transmission, which increases complexity and reduces reliability. The proposed approach employs a dual-input, dual-output wireless power transfer system with anti-parallel outputs and a self-driven MOSFET–diode configuration, enabling bipolar current flow without rotor-side active converters or control signals. This simplifies the system, reduces control complexity, and enhances reliability. A small-scale prototype has been experimentally validated, demonstrating bipolar current reversal up to 2.88 A (118 W) with a self-driven rectifier efficiency of 99% under an inductive load emulating the field windings.

Index Terms—pole-changing, electrically excited synchronous machine, field excitation, wireless power transfer

I. INTRODUCTION

Electrically excited synchronous machines (EESMs) provide a rare-earth-free alternative to permanent-magnet synchronous machines (PMSMs) for traction applications by offering high torque density and controllable field excitation [1], [2]. However, conventional EESMs have a fixed torque–speed characteristic determined at the design stage, often resulting in suboptimal efficiency under varying operating conditions. Pole-changing techniques address this limitation by enabling adjustment of the pole number, as illustrated in Fig. 1.a, which modifies the torque–speed profile to better match the operating point [3], [4]. This approach might improve efficiency across a wide speed range, which is particularly beneficial for traction drives [5].

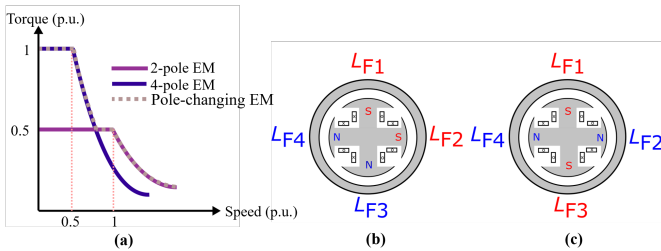


Fig. 1. (a) Torque–speed characteristics of 2-pole, 4-pole, and pole-changing EESMs, along with corresponding rotor winding configurations and field current directions. (b) 2-pole EESM configuration. (c) 4-pole EESM configuration.

Most prior work has focused on pole-changing induction machines (IMs), where altering the stator pole number inherently results in a corresponding change in the rotor [6]. In contrast, pole-changing EESMs require reconfiguration of the rotor poles, which is shown in Fig. 1.b and Fig. 1.c as an example, making the excitation system a critical element [7].

Further, the rotor excitation in EESMs is complicated on its own, and several solutions have been proposed. The most common solution is to use slip rings, which consist of conductor rings and carbon brushes. While simple and cost-effective, this method suffers from reliability concerns due to wear and fatigue, particularly at high speeds [8]. An alternative is to use contactless field excitation (CFE) systems based on wireless power transfer (WPT), as exemplified in Fig. 2, which eliminates mechanical contact and thereby reduces friction and maintenance requirements [9].

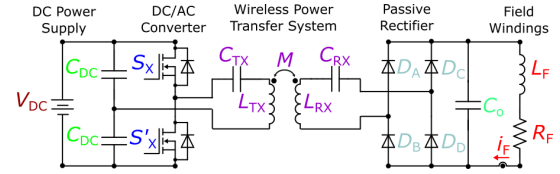


Fig. 2. A contactless field excitation system employing a wireless power transfer system based on series-series compensated inductive coupling.

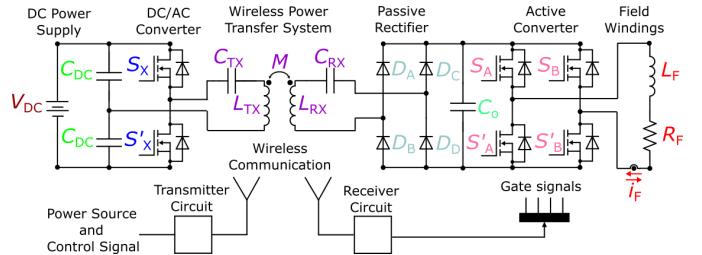


Fig. 3. A hypothetical bipolar current-capable wireless power transfer system based on series-series compensated inductive coupling. The system utilizes an active converter on the rotating side, with wireless communication used to transmit gate-drive signals.

However, implementing WPT-based CFE systems in pole-changing machines presents additional challenges, since rotor excitation requires polarity reversal of the field current. A conceptual implementation, shown in Fig. 3, necessitates an additional active converter to realize this functionality, along with corresponding control signal transmission to the rotor side.

To address this issue, several self-driven or brushless approaches have been proposed to achieve bipolar current without active converters. Dey et al. [10] and Jiang et al. [11] employed dual resonant coils with receiver-side MOSFETs triggered by pulse transformers to achieve electrically/hybrid-excited synchronous motors and wireless servo motors, respectively. Huang et al. [12] proposed a wireless motor based on a three-phase six-stator-winding PMSM. This system has redundant stator windings, which employ a multi-branch WPT system, to achieve bipolar current. In contrast, this letter achieves bipolar excitation using anti-parallel rectifiers with

a simple diode–MOSFET pair, thereby eliminating active converters, pulse transformers, rotor-side controllers, and redundant windings, while significantly simplifying the system.

II. THE PROPOSED SYSTEM STRUCTURE

To achieve bipolar current capability, the proposed system departs from the conventional single-input, single-output WPT architectures and adopts a dual-input, dual-output configuration, illustrated in Fig. 4.

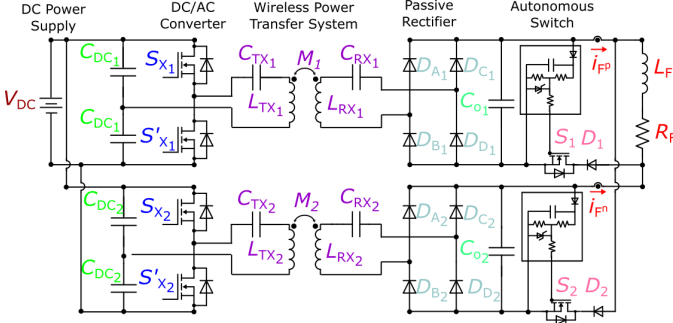


Fig. 4. Proposed bipolar dual-input dual-output contactless field excitation system based on WPT.

The proposed dual-input, dual-output WPT system employs two resonant coil sets, each feeding a diode rectifier to supply unidirectional current. The anti-parallel connection of the rectifier outputs enables bipolar excitation of the field winding. To avoid reverse biasing of the rectifiers, bidirectional power switches—implemented using a MOSFET in anti-series with a diode—are placed between the rectifier ground and one terminal of the field winding. These switches are self-driven using passive feedback network from their respective rectifier outputs, eliminating the need for external control signals. Field-current reversal is achieved by altering excitation between the two coil sets: one set drives the positive direction, while the other drives the negative.

III. SYSTEM DESIGN

A. Dual-Input Dual-Output WPT System Design

A series–series (SS) compensated topology is adopted for the proposed WPT system, since its resonant frequency is independent of coupling and load variations. Although the system features a dual-input, dual-output configuration, the design procedure is equivalent to that of a single-input, single-output system, since each transmitter is driven by an independent converter and cross-coupling can be neglected. Following the systematic method in [13], the final circuit parameters are obtained as follows.

The reflected load resistance is

$$R_{F-ref} = \frac{8}{\pi^2} R_F, \quad (1)$$

where R_F is the field resistance and the receiver inductances are given by

$$L_{RX1,2} = \frac{Q_{RX1,2} R_{F-ref}}{\omega_0}, \quad (2)$$

where ω_0 is the resonant frequency and $Q_{RX1,2}$ is the chosen quality factor. Typical values of $Q_{RX1,2}$ (2–10) ensure a trade-off between coil size and selectivity.

The transmitter inductances are derived from the coupling coefficients $k_{1,2}$ and mutual inductances $M_{1,2}$ as

$$L_{TX1,2} = \frac{M_{1,2}^2}{k_{1,2}^2 L_{RX1,2}}, \quad (3)$$

where $k_{1,2}$ are selected to avoid bifurcation according to [13]. The resonant capacitors follow directly as

$$C_{TX1,2} = \frac{1}{L_{TX1,2} \omega_0^2}, \quad C_{RX1,2} = \frac{1}{L_{RX1,2} \omega_0^2}. \quad (4)$$

The resonant frequency ω_0 can be chosen within a practical range to balance system efficiency and component size, while the output voltage can be regulated either by slight frequency detuning or by duty-cycle control of the DC/AC converters.

B. Feedback Network (Autonomous Gate Driver)

The feedback network in Fig. 5 autonomously drives the MOSFET on the receiver side.

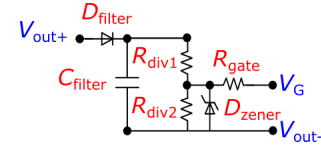


Fig. 5. Autonomous gate-drive circuit implemented on the secondary side.

The rectified outputs (V_{out+} , V_{out-}) are filtered by a capacitor-diode pair (C_{filter} , D_{filter}) and scaled down through a resistive divider (R_{div1} , R_{div2}) to bias the gate–source voltage of the MOSFET. The divider ratio is selected such that the MOSFET turns on at the minimum field current, given by

$$V_{GSmin} = I_{Fmin} R_F \left(\frac{R_{div2}}{R_{div1} + R_{div2}} \right), \quad (5)$$

where I_{Fmin} is the minimum field current and R_F is the field winding resistance. To protect the MOSFET gate at high field currents (i.e. higher rectified voltages), a Zener diode (D_{zener}) is utilized to clamp the gate–source voltage ($V_{GS} = V_G - V_{out-}$), while a series gate resistor (R_{gate}) is included to suppress ringing.

IV. EXPERIMENTAL VALIDATION

An experimental setup was constructed to validate the proposed system, as shown in Fig. 6.

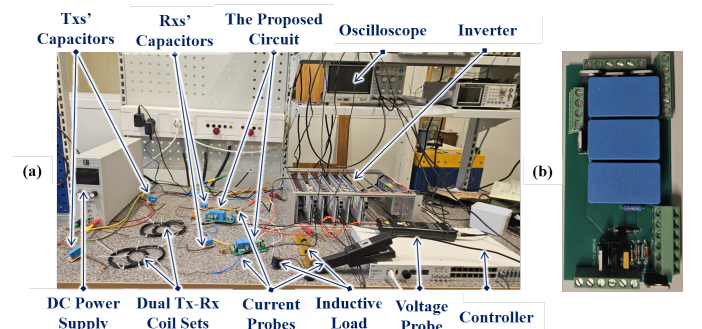


Fig. 6. Experimental setup of the proposed system. a) Overall view of the setup. b) Printed circuit board (PCB) of the rectifier circuit with a self-driven switch.

The corresponding circuit parameters and component ratings are summarized in Table I. In the absence of the pole-changing EESM, the field winding was emulated by an inductive load as a proof of concept.

TABLE I
CIRCUIT COMPONENTS, RATINGS, AND SYSTEM PARAMETERS

Component	Model (Brand)	Rating / Value
Diode (Rectifier/Filter)	STPS20H100CT (STMicroelectronics)	100 V, 2x10 A
Zener Diode	1N5341BG (Onsemi)	6.2 V, 5 W
MOSFET (Autonomous Switch)	IRFI540NPBF (Infineon)	100 V, 20 A
Diode (Autonomous Switch)	STPS20H100CT (STMicroelectronics)	100 V, 2x10 A
Resistor (Divider)	Generic	5 k Ω , 20 k Ω
Capacitor (Filter/Output)	Generic	100 nF, 50 μ F
Inverter Module	PEB8024 (Imperix)	
Controller	B-Box RCP (Imperix)	

WPT System Parameters

Tx Coils	110 μ H
Rx Coils	40 μ H
Tx Capacitors	26 nF
Rx Capacitors	66 nF
Mutual Inductance	12.5 μ H
Resonant Frequency	95kHz
Operating Frequency	95kHz-110kHz
Input Voltage	72V
Load (Field) Inductance	600 μ H
Load (Field) Resistance	14.2 Ω

A. Verification of the Circuit Operation

In this experiment, only one coil set was excited, resulting in unidirectional current. Fig. 7 shows the measured waveforms of the rectifier input voltage, the MOSFET gate-source voltage, and the resulting field current.

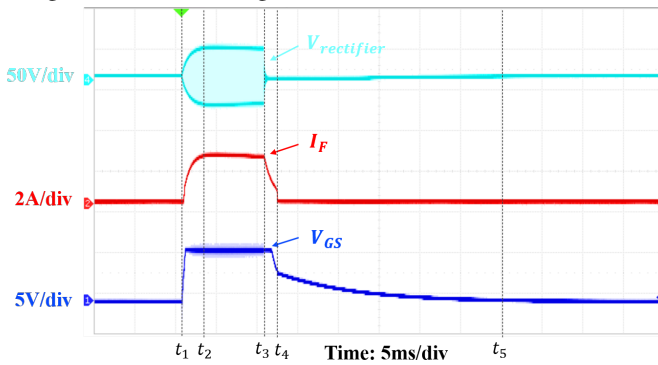


Fig. 7. Measured waveforms of the rectifier input voltage, gate-source voltage, and field current during unidirectional operation of the proposed system.

The rectifier output exhibits the expected pulsating waveform of high-frequency WPT systems, which biases the MOSFET gate-source via the feedback network. The sequence of operation is as follows:

- At t_1 , excitation begins; the rectifier input voltage rises, and the gate-source voltage (V_{GS}) exceeds the threshold, turning on the MOSFET and initiating the rise of the field current (I_F).

- At t_2 , I_F reaches steady state, while V_{GS} is clamped by the Zener diode.
- Between t_2 and t_3 , steady-state operation is maintained with constant I_F and clamped V_{GS} .
- At t_3 , excitation ceases; the rectifier voltage collapses, and I_F decays toward zero while V_{GS} remains above the threshold.
- At t_4 , V_{GS} falls below the threshold, turning off the MOSFET; the remaining field current dissipates via the field resistance.
- Between t_4 and t_5 , V_{GS} continues to discharge through the feedback network until fully depleted.

These results confirm that the proposed self-driven mechanism enables proper excitation without the need for external gate drivers or control signals.

B. Verification of Current Reversal

In this experiment, the bipolar capability of the proposed excitation system was validated by alternately energizing the two transmitter-receiver coil sets. Fig. 8 presents the measured receiver currents of each coil set and the resulting field current.

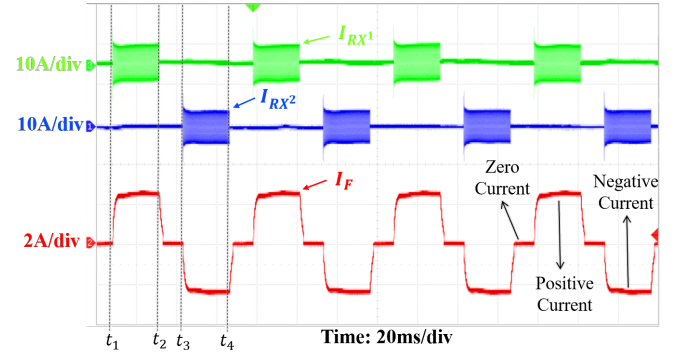


Fig. 8. Measured waveforms of receiver currents and resulting field current during bipolar operation of the proposed contactless excitation system.

The activation sequence is summarized as follows:

- At t_1 , the first coil set is energized, producing a positive rectified voltage and a positive field current.
- At t_2 , the first coil set is de-energized, and the current decays toward zero.
- Between t_2 and t_3 , a blanking interval occurs with zero field current before the second coil set is excited.
- At t_3 , the second coil set is energized, producing an opposite rectified voltage and a negative field current.
- At t_4 , the second coil set is de-energized, and the current decays to zero.

These results confirm that the proposed system achieves bipolar excitation without active control or signal transmission on the receiver side. In the present implementation, current decay occurs passively. First, the field current free-wheels to zero, then a blanking interval is introduced. The free-wheeling time is dictated by the field winding time constant ($\tau_f = L_f/R_f$) and cannot be actively controlled; however, the blanking interval, introduced as a safety margin, can be reduced provided that zero field current is ensured. Nevertheless, in traction drives, the resulting short no-torque interval, comprising both free-wheeling and blanking time, can be scheduled like a gear shift so that the vehicle inertia attenuates the transient without

noticeable speed dips, as similar torque interruptions have also been reported in the pole-changing machine literature [14].

C. Field Current Regulation Method

Field current regulation is essential for achieving maximum torque-per-ampere (MTPA) operation and enabling field-weakening modes in EESMs. In the proposed system, the field current is controlled by detuning the WPT operating frequency from resonance. Fig. 9 presents the transient responses of the field and transmitter currents during frequency detuning and retuning.

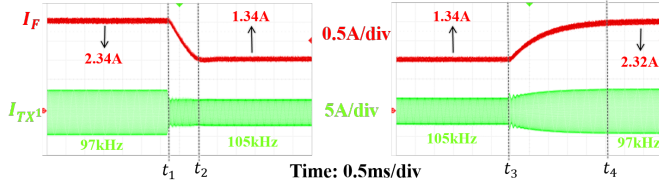


Fig. 9. Measured transmitter and field currents showing excitation current reduction during frequency detuning and increase during frequency retuning.

The sequence of the operations is summarized as follows:

- Before t_1 , the system operates near resonance at 97 kHz (resonant frequency is 95 kHz), with a field current of ≈ 2.34 A.
- At t_1 , the operating frequency is increased to 105 kHz, detuning the WPT system.
- At t_2 , a new steady-state is reached with a reduced field current of ≈ 1.34 A.
- At t_3 , the frequency is shifted back toward 97 kHz, returning to operation near resonance.
- At t_4 , the field current reaches steady-state at ≈ 2.32 A.

These results confirm that smooth field-current control can be achieved through operating-frequency adjustment while maintaining stable autonomous gate-driver operation without oscillatory transients.

D. Efficiency Performance under Varying Load Currents

The efficiencies of the proposed system, measured from the rectifier input to the load under varying load currents, are summarized in Table II.

TABLE II
MEASURED EFFICIENCY OF THE PROPOSED SYSTEM

Load Current (A)	2.88	2.21	1.88	1.47	0.612
Efficiency (%)	99.0	97.8	96.9	96.4	87.0

As expected, efficiency improves with load current, since the diode voltage drop becomes less significant relative to the output voltage. At light load (0.61 A), it reduces efficiency to about 87%, which is a known limitation of diode-rectified WPT systems. Compared with conventional WPT exciters, the proposed design introduces only one additional series diode-MOSFET pair, resulting in negligible extra loss. Consequently, the autonomous switching concept maintains high efficiency (up to 99%) while enabling field-current reversal.

V. CONCLUSION

This work proposes a novel bipolar field excitation system for pole-changing electrically excited synchronous machines. The system utilizes a dual-input dual-output wireless power

transfer system that enables bipolar field current without requiring active converters, pulse transformers, transmission of wireless control signals to the receiver side, or redundant field windings. By introducing anti-parallel diode rectifiers at the output with a self-driven switch, current reversal is achieved. A small-scale prototype has been experimentally validated, achieving current reversal up to 2.88 A (118 W) with a self-driven rectifier efficiency of 99%. The approach offers simplified design, reduced control complexity, and eliminates the need for active circuits on the rotating side. Future work will scale the concept to traction-level EESM rotor windings, addressing field-current dynamics, thermal management, and scalability, as well as exploring stator-harmonic-based excitation as an alternative to the WPT link.

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